

1CK60 Maintenance and Service Logistics

Assignment 1

Fall 2025

Please note

- The grade for this assignment determines 25% of your final grade.
- For every question the number of points that can be obtained is given.
- You can work on this assignment in a group of at most two students. Put your name and that of your teammate at the beginning of your report. You also need to register together as a group on Canvas. (We don't recommend to work on the assignment alone.)
- You should upload your report and source code to Canvas together in one .zip file. The deadline is Tuesday 2 December, 23:59 (CET).
- We ask you to enter the following information in the python file: name and surname. If you decide to perform the assignment alone, then in the Python template set the variables for the second student as follows: `student2.name="empty"`, and `student2.surname="empty"`.
- For all questions you need to implement evaluation and optimization procedures. You are required to do this in the file "Template_1CK60_MSL_2025_2026_Assignment_1.py". The programming language we expect you to use is Python (version 3.11.11 or any version that is higher).
- The Python packages you need to use to complete the exercises are given in the template. If you use a built-in function, please make explicit which function you import as demonstrated below.
 - Good: `from math import factorial, floor`
 - Bad: `from math import *`
 - Bad: `import math`
- For those who have limited experience with programming or Python, a lecture will be put online with some programming examples to learn the basics of Python.
- Your code should be platform-independent. For instance, there should be no file paths in your code, e.g., `"C:\Users\..."`.
- The file `Auto.Grader_1CK60_MSL_2025_2026_Assignment_1.py` will be used to grade your solutions to the programming questions. For these questions, you will only get points based on the output of your program. If

your program does not completely execute, you will not get points for the programming exercises. For instance, no points are given for programs that crash or end up in an infinite loop.

- Check if you have written your name and surname in the Python file.
- If you decide not to make a specific exercise, then please leave the function as it is in the file. This will prevent errors.
- You are expected to use the report template (Report_Template_1CK60_MSL_2025_2026_Assignment_1.docx) for answering report questions. You should convert this template to pdf for submission.
- DO NOT change the names of the provided functions.
- DO NOT change the input of the provided functions.
- DO NOT change the order of variables in the input of the provided functions.
- DO NOT round your output.
- Your code will be graded with a set of test cases that is different than the one provided. So make sure your code works with different inputs and with different numbers of SKUs.
- We grade what we have received at the deadline. Don't wait until the deadline to upload your work; if something goes wrong, you have a 0 grade.
- We can schedule a meeting with each member of the group to see if that member truly understands everything that is handed in (which includes both code and report). Grades may be adapted based on this meeting.

To do before handing in the assignment

You may use any programming or integrated development environment that you like. If you would use Spyder:

- Restart Spyder or select the remove all variables option.¹ Then check if your code runs in the Spyder.
- Restart Spyder or select the remove all variables option. Put your file in the 'Submissions' folder in the 'Grading_set_up' folder. Then run the file Auto_Grader_1CK60_MSL_2025_2026_Assignment_1.py and check if it grades your assignment.

¹This is to ensure that the console is empty before you run your code. If the console is not empty, it might happen you have some variables defined from a previous version of your program that affects the output of your current code.

Introduction

For this assignment you will implement several methods to analyse and optimize spare part inventories. Exercises are divided into two groups of questions: programming questions and report questions. You are required to make the programming questions in the file “Template_1CK60_MSL_2025_2026_Assignment_1.py”. In this file there is one function for each programming question. This function name starts with the number of the exercise. To answer the corresponding exercise complete the function. You are allowed to change the name of the file, but we ask you not to change the names, the inputs and the order of the inputs of the provided functions. Furthermore, the output of each function should be as specified in the Python template file. However, you are allowed to define your own additional functions to help you solve the questions.

Use the template for answering the report questions Report_Template_1CK60_MSL_2025_2026_Assignment_1.docx. You should hand in as a PDF file. The questions that should be answered in the report are indicated with **(report)** in front of them. We encourage you to numerically motivate your answers on report questions unless stated otherwise. So if your answer is that one solution has a lower waiting time than another solution, give the waiting times you calculated for both solutions. We ask you to hand in both the .py file and your report together by putting them in a .zip file.

You receive two different data sets to use throughout this assignment.

- “Data_Q1_Q3_Q4.csv”: contains the variables of 15 SKUs and also three base stock levels per SKU. This data is used for **Q1**, **Q3**, and **Q4**.
- “Data_Q2.csv”: contains on-hand stock levels at the beginning of 36 months for 50 SKUs. This data is used for **Q2**.

For the programming exercises of **Q1**, **Q2**, **Q3** and **Q4** the programming template reads in the relevant data from the Excel files and outputs your functions with this data as input. It might be that you need to output other results to motivate your report questions. The Python template has code that takes as input the data from the Excel file and outputs it with the functions you have implemented.

Good luck with the assignment!

Questions (100 points)

To help you answer the questions, we suggest you study Chapter 2 of Van Houtum and Kranenburg (2015).

- (Q1) Stark Technologies wants to improve its spare parts management. They identified fifteen stock keeping units (SKUs) S_i $i \in I = 1, 2, 3, \dots, 15$ of which the demand follows an independent Poisson process and the demand satisfies the other three conditions in van Houtum and Kranenburg (2015). The file Data_Q1-Q3-Q4.csv contains three base stock policies $\mathbf{S}^A$, $\mathbf{S}^B$, $\mathbf{S}^C$ in columns B, C, and D, the acquisition costs c_i^a (column E), the total expected number of failures per year m_i (column F), and the average repair lead time in years t_i (column G) for all SKU $i \in I$. For this question, the company assumes backordering.

For the subquestions (a) to (i), implement the functions in the specific place in the given template. Please note that these functions should be able to calculate the measures for different inputs. To calculate the measures for three base stock policies with given inputs in Q1, use the area in the template (SOLUTION QUESTION 1) that prints the calculated values using the functions you have implemented.

- (a) (3 points) Implement the function for the expected number of backorders $EBO_i(S_i)$ of SKU $i \in I$.
 - (b) (3 points) Implement the function for the sum of expected back orders for base stock policy $\mathbf{S}$ $EBO(\mathbf{S})$.
 - (c) (3 points) Implement the function for the backorder probability $PBO_i(S_i)$ of SKU $i \in I$.
 - (d) (3 points) Implement the function for the sum of backorder probabilities $PBO(\mathbf{S})$.
 - (e) (3 points) Implement the function for the total acquisition cost $C(\mathbf{S})$.
 - (f) (3 points) Implement the function for the fill rate $\beta_i(S_i)$ of SKU $i \in I$.
 - (g) (3 points) Implement the function for the aggregate fill rate $\beta(\mathbf{S})$.
 - (h) (3 points) Implement the function for the waiting time $W_i(S_i)$ of SKU $i \in I$.
 - (i) (3 points) Implement the function for the aggregate mean waiting time $W(\mathbf{S})$.
 - (j) (6 points) (**report**) Calculate the measures from (a) to (j) in the SOLUTION QUESTION 1 part of the code by calling the functions you have implemented. Based on these calculations for three base stock policies, fill the report template and discuss which base stock policy should be chosen.
- (Q2) Tyrell Parts has 50 SKUs for which the managers want to determine base stock levels. The data for these SKUs are provided in Data_Q2.csv file.

Column A represents the SKU $i \in I$, $I = 1, 2, \dots, 50$, column B gives the acquisition costs, column C gives the repair lead times, column D gives the average demand rate that is currently used by the company, columns F to AO give the demand for each of the previous 36 months. Column E will give your improved estimate of the average demand per month. For this question, the company assumes backordering.

For the subquestions (a) and (b), implement the functions in the specific place in the given template. Please note that these functions should be able to calculate the base stock levels for different inputs. To calculate the base stock levels with given inputs in Q2, use the area in the template (SOLUTION QUESTION 2) that prints the calculated base stock levels using the functions you have implemented.

- (a) (2 points) **(report)** Before using optimization techniques to set spare part base stock levels, it is important to check that data does not contain errors. Indicate in the report template the errors you found. For each error, indicate in your report the row and column number where the error occurs. Do not look for errors in the demand rate columns (D and E), yet.
Example: the cost of SKU number 1 is in position 2B.
- (b) (2 points) **(report)** You have identified the errors in the data. The company wishes to re-evaluate the average demand of each SKU. Put a 1 on the location of all values with error in Data_Q2.csv that you found in part (a). Complete the function 'Q2b_NewDemandRates' to find the average demand rate over all periods. If you run your code, this will also put your demand rate estimates in the column E of the .csv file. What original average demand rates (column D) differ more than 1 from your estimated average demand rates (column E)? Indicate the row numbers in your report template (not the SKU numbers).
- (c) (5 points) Implement the greedy algorithm for setting base stock levels such that $EBO^{obj} = 0.5$ is satisfied.
- (d) (5 points) Implement the item approach for setting base stock levels such that $EBO^{obj} = 0.5$ is satisfied.
- (e) (6 points) **(report)** Divide the SKUs in two groups as the first 25 SKUs (rows 2 to 26), and the second 25 SKUs (rows 27 to 51). Use the demand rates you found in part (b). Find the base stock levels with greedy algorithm (c) and the item approach (d) for these two groups of SKUs using the area in the template (SOLUTION QUESTION 2) by calling the functions you have implemented in Q1. Based on these calculations, fill the report template and discuss the differences

between the base stock levels, $EBOs$ and $C(S)$ s found with these two methods for two groups of SKUs.

- (Q3)** Stark Technologies (from Q1) now considers using an item approach to ensure that the stock levels are such that the aggregate mean waiting time satisfies $W^{obj} = 0.05$ for the fifteen SKUs in the Data_Q1_Q3_Q4.csv file. The motivation of the company is that the time spent waiting determines how much downtime there is instead of the number of backorders. They want to be able to set individual waiting times and implement an item approach for the mean waiting time. They hired three consultants each of which has a different proposal on how to set the mean waiting time constraint for each SKU $i \in I$. For this question, the company assumes backordering.

For the subquestions (a) to (c), implement the functions in the specific place in the given template. Please note that these functions should be able work with different inputs. To calculate the costs in Q3, use the area in the template (SOLUTION QUESTION 3) that prints the costs using the functions you have implemented.

- (a) (4 points) Consultant 1 suggests setting $W_i^{obj} = \frac{m_i}{M} W^{obj} = \frac{m_i}{114} 0.05$ for every SKU $i \in I$, since the mean waiting times should be aggregated just like the targets for the item approach with the expected backorders. Implement the item approach with the method that Consultant 1 suggests.
- (b) (4 points) Consultant 2 suggests setting $W_i^{obj} = 0.05$ for every SKU $i \in I$, since each individual SKU needs to satisfy the mean waiting time constraint. Implement the item approach with the method that Consultant 2 suggests.
- (c) (4 points) Consultant 3 suggests setting $W_i^{obj} = 0.05/M$ where $M = \sum_{i \in I} m_i = 114$ for every SKU $i \in I$, since the mean waiting time objective needs to be spread out among different parts. Implement the item approach with the method that Consultant 3 suggests.
- (d) (6 points) **(report)** Find the base stock levels and calculate the aggregate mean waiting time and the annual costs for three consultants in the SOLUTION QUESTION 3 part of the code by calling the functions you have implemented in. Based on these calculations, fill the report template and compare the waiting times and the annual costs found by the methods suggested to help the manager for the decision she should make.
- (e) (7 points) **(report)** As argued in Section 2.7.1 in van Houtum and Kranenburg (2015), the constraint on the aggregate mean waiting

time W^{obj} can be transformed into a constraint on the expected back-orders EBO^{obj} . Which of the methods suggested by the consultants agrees (finds the same solution for every set of input parameters) with the item approach for $EBO(S)$ given in Section 2.6 in van Houtum and Kranenburg (2015)? Provide an analytical motivation for your answer in the report template.

- (Q4) Stark Technologies (from Q1 and Q3) decides to have another plant in a different country. The processes will be similar with similar SKUs. However, they want to satisfy $W^{obj} = 0.001$ for the fifteen SKUs in the Data_Q1_Q3_Q4.csv file. since having downtimes will be more costly at the new location.

For the subquestions (a) to (d), implement the functions in the specific place in the given template. Please note that these functions should be able work with different inputs. To calculate the costs in Q4, use the area in the template (SOLUTION QUESTION 4) that prints the costs using the functions you have implemented.

- (a) (4 points) Implement the greedy algorithm to find the base stock levels such that the $W^{obj} = 0.001$ is satisfied. Use the annual holding costs c_i^h in column H of the Data_Q1_Q3_Q4.csv file for implementing the algorithm.
- (b) (4 points) The team suggests shipping parts from the original country with additional shipment cost (column I) since the shipping time (column J) is shorter compared to repair lead times. Implement the cost function $\hat{C}(S)$ to calculate the total costs with the shipping option.
- (c) (4 points) Implement the function for the aggregate mean waiting time $W(S)$ with the shipping option.
- (d) (4 points) Implement the greedy algorithm to find the base stock levels with the shipping option such that the given waiting time target $W^{obj} = 0.001$ is satisfied.
- (e) (6 points) **(report)** Find the base stock levels and calculate the annual costs for (a) and (d) by calling the functions you have implemented. Based on these calculations, fill the report template and compare the annual costs with and without using shipping option to help the company for the decision to be made.

References

Van Houtum, G.J. and Kranenburg, B. (2015). *Spare parts inventory control under system availability constraints*, volume 227. Springer.